

GeoMMBench and GeoMMAgent: Toward Expert-Level Multimodal Intelligence in Geoscience and Remote Sensing

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Abstract

Recent advances in multimodal large language models (MLLMs) have accelerated progress in domain-oriented AI, yet their development in geoscience and remote sensing (RS) remains constrained by distinctive challenges: wide-ranging disciplinary knowledge, heterogeneous sensor modalities, and a fragmented spectrum of tasks. To bridge these gaps, we introduce GeoMMBench, a comprehensive multimodal question-answering benchmark covering diverse RS disciplines, sensors, and tasks, enabling broader and more rigorous evaluation than prior benchmarks. Using GeoMMBench, we assess 36 open-source and proprietary large language models, uncovering systematic deficiencies in domain knowledge, perceptual grounding, and reasoning—capabilities essential for expert-level geospatial interpretation. Beyond evaluation, we propose GeoMMAgent, a multi-agent framework that strategically integrates retrieval, perception, and reasoning through domain-specific RS models and tools. Extensive experimental results demonstrate that GeoMMAgent significantly outperforms standalone LLMs, underscoring the importance of tool-augmented agents for dynamically tackling complex geoscience and RS challenges. Project: <https://geo-mm-agi.github.io>

1. Introduction

Recent progress in multimodal large language models (MLLMs) has accelerated the pursuit of artificial general intelligence (AGI), marking an important milestone with the emergence of Expert AGI. This level of machine intelligence is characterized by performance comparable to or exceeding that of skilled human experts at the 90th percentile across professional domains [31]. As research advances toward this stage, a growing collection of multimodal benchmarks has been introduced to assess reasoning, perception, and domain expertise across diverse disciplines [16, 20, 25, 50, 52]. These benchmarks serve as es-

sential instruments for measuring progress and guiding the development of more capable multimodal systems.

Despite these advances, the domain of geoscience and remote sensing (RS) has received limited attention in current benchmarking efforts. Existing studies [8, 18, 19, 38, 43] remain narrow in scope, focusing primarily on perception tasks such as classification, detection, or segmentation, and are often restricted to a limited set of sensor modalities, particularly optical imagery. In contrast, geoscience requires the integration of multi-sensor data, spatiotemporal reasoning, and multidisciplinary knowledge, which existing benchmarks fail to adequately evaluate. This gap highlights the need for a comprehensive and expert-level assessment framework to systematically measure and advance AGI performance within the RS domain.

To bridge this gap, we introduce **GeoMMBench**, a comprehensive domain-specific benchmark designed to evaluate the multidisciplinary and multimodal capabilities of MLLMs in geoscience and RS. The benchmark contains 1,053 expert-level, image-based multiple-choice questions that represent diverse and realistic challenges across the key dimensions, as illustrated in Fig. 1.

GeoMMBench is characterized by three major properties that define its breadth and depth. (1) *Multi-disciplinary coverage*: It spans multiple geospatial disciplines, including RS, photogrammetry, geographic information systems (GIS), and global navigation satellite systems (GNSS). These areas draw upon foundational knowledge in mathematics, physics, cartography, spectroscopy, and geography, mirroring the diverse expertise required for professional interpretation and decision-making. (2) *Multi-sensor modalities*: The benchmark incorporates a wide range of sensing sources such as optical imagery, synthetic aperture radar (SAR), multispectral and hyperspectral data, LiDAR, and digital elevation models (DEM). These complementary modalities enable a realistic evaluation of multi-sensor integration skills crucial for expert-level analysis. (3) *Diverse task spectrum*: GeoMMBench covers a continuum of complexity, from theoretical and data-preprocessing concepts to perception tasks and high-level geospatial applications, providing a holistic assessment of AGI capabilities in RS.

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We conduct a comprehensive evaluation of 36 open-source and proprietary MLLMs on GeoMMBench, revealing critical gaps in their ability to interpret geoscience data. These limitations stem from the benchmark’s multidimensional structure, high domain complexity, and requirement for expert-level reasoning. To address these challenges, we propose **GeoMMAgent**, a multi-agent system that advances MLLMs toward expert-level performance in geoscience and RS. GeoMMAgent integrates large language models (LLMs) with domain-specific tools and models within an adaptive framework that plans, coordinates, and executes tasks across three key functional stages: retrieval, perception, and reasoning. Through strategic task decomposition and tool selection, GeoMMAgent enables models to access external knowledge bases, analyze complex RS data, and perform spatial reasoning beyond the intrinsic capacity of general-purpose MLLMs. The framework supports transparent and modular integration of various RS tools and model components, promoting interpretability, scalability, and future extensibility.

Our main contributions are summarized as follows. *First*, we propose GeoMMBench, a comprehensive and domain-specific benchmark, which evaluates the multidisciplinary, multimodal, and multi-level reasoning capabilities of MLLMs in geoscience and RS. *Second*, we develop GeoMMAgent, a multi-agent system tailored to geoscience and RS. It integrates MLLMs with domain-specific models and tools through structured task planning that spans retrieval, perception, and reasoning stages, enabling adaptive coordination and expert-level task execution. *Third*, we conduct an extensive evaluation of open-source and proprietary MLLMs on GeoMMBench, establishing the current performance landscape, identifying key challenges in domain expertise, and demonstrating the effectiveness of GeoMMAgent in bridging these capability gaps.

2. Related Works

MLLMs for Geoscience and Remote Sensing. Despite rapid progress in MLLMs, their applications in geoscience and RS remain limited [48, 49]. Most recent RS studies focus on VQA over optical RGB imagery [30, 59]. GeoChat [19] fine-tunes LLaVA-1.5 [23] with detection-based instruction data, while TeoChat [18] adapts temporal Earth observation data for change detection. LHRs-Bot [32] and EarthVQA [43] further tailor multimodal datasets for RS perception and pixel-level understanding. However, these models remain task-specific and lack multidisciplinary knowledge integration, multi-sensor awareness, and advanced reasoning, which are key capabilities for expert-level intelligence in geoscience and RS.

Knowledge-driven Benchmarks for RS. Progress toward Expert AGI [31] has inspired knowledge-oriented benchmarks to assess professional expertise. Early textual eval-

Disciplines	Photogrammetry (130)		
	RS (142)	GIS (46)	GNSS (87)
Sensor Modalities	Optical (215)		LiDAR (64)
	MSI/HSI (171)		SAR (163)
	Thermal (72)		DEM (163)
Task Spectrum	Data Quality Inspection (28)		
	Theoretical knowledge (394)		
	Image Perception (307)		
	Spatial Relation (40)		
	Time-series Analysis (48)		
	Geospatial Applications (125)		

Figure 1. Expert-level knowledge dimensions in geoscience and RS covered in GeoMMBench.

uations such as AGIEval [58], ARC [4], and MMLU [13] focus on standardized exams, while MMLU-Pro [46] introduces higher-order reasoning. Multimodal extensions like ScienceQA [27], MMMU [52], MMMU-Pro [53], and Video-MMMU [15] broaden evaluation across images and temporal sequences. In contrast, RS benchmarks still concentrate on perception tasks such as scene classification [6, 28, 47, 51], object detection [22, 37], and semantic segmentation [22, 43], often using automatically generated descriptions instead of expert assessments. To the best of our knowledge, GeoMMBench is the first expert-crafted, knowledge-driven benchmark that spans multiple disciplines, sensor modalities, and task hierarchies, bridging theoretical concepts and practical applications in geoscience and RS.

3. GeoMMBench Benchmark

3.1. Overview

GeoMMBench is a rigorously designed benchmark that evaluates the ability of MLLMs to comprehend geoscience and RS concepts across multiple dimensions. Its primary objective is to assess how effectively these models perceive and interpret geospatial information across related disciplines while applying domain-specific knowledge and reasoning to solve complex geospatial problems.

The benchmark comprises 1,053 image-based multiple-choice questions, each containing a single correct answer derived from the provided imagery. The dataset was developed through collaborative efforts by domain experts in geoscience and RS, including PhD researchers and doctoral students, who engaged in extensive panel discussions to define the scope, content, and evaluation dimensions. Source materials were drawn from a diverse range of authoritative references, including educational resources, online repositories, and academic literature.

For evaluation purposes, the dataset is divided into a *val* set of 37 questions and a *test* set of 1,016 questions. The *val* set is intended for assessing human expert performance and for model selection or hyperparameter tuning in future

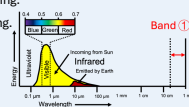
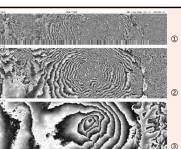
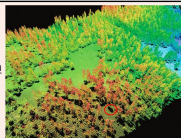
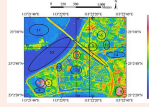
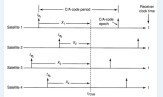
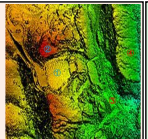
Different Disciplines	Different Sensor Modalities	Different Tasks
Which of the following statements about Band ① is accurate? A) It is often used in passive remote sensing. B) It is suitable for thermal remote sensing. C) It operates only during the day and is affected by weather conditions. D) It can penetrate vegetation to detect subsurface targets.  Discipline: RS; Image Type: Diagram	Which SAR bands correspond to interferograms ①, ②, and ③, respectively? A) X, C, L B) X, L, C C) C, X, L D) C, L, X.  Data Modality: SAR; Image Type: interferogram image	Which of the following inferences is most likely correct based on the image? A. Wetland converted to lake B. River expanded to lake C. Saline-alkali or sandy land converted to lake D. Glacier meltwater became lake  Task: Time Series Analysis; Image Type: Map, Optical image
Which step in photogrammetry is illustrated in the image? A) Super-resolution B) Motion compensation C) Radiometric correction D) Stereo image matching  Discipline: Photogrammetry; Image Type: Panchromatic Image	What does the image depict? A) A thermal image of a forest B) A LiDAR point cloud of forest vegetation C) A LiDAR point cloud of urban areas D) A simulated 3D model of a grass field  Modality: LiDAR; Image Type: LiDAR image	Does any quality issue appear in this remote sensing image? A. Twisted Object B. Blur C. Stitching Misalignment D. Strip Noise  Task: Image Quality Inspection; Image Type: Optical image
In the GIS project illustrated, which spatial query is most appropriate for determining the number of villages located near the highway? A) Select villages that intersect the buffer zone. B) Find all points that have a spatial join. C) Select point-in-polygon query. D) Analyze the distance calculation.  Discipline: GIS; Image Type: Map	Which of the following statements about the image is NOT correct? A) The river (No. 10) registered a much lower temperature than both lakes (Nos. 3 and 4). B) Lakes (Nos. 8 and 9) show low land surface temperature regions on the map. C) There is a temperature difference between the central areas of both football grounds (Nos. 3&4) due to their natural and artificial lawns. D) In the residential area (No. 11), the average land surface temperature reaches up to 60°C.  Modality: Thermal; Image Type: Thermal image	In which direction from the hospital (marked by red box) is the parking lot (blue box) located? A) East B) Northeast C) Northwest D) West  Task: Spatial Relation Analysis; Image Type: Optical Image
Let c be the speed of light, and c_r be the C/A-code chipping rate. How is the pseudorange to a satellite calculated based on the given figure? A) $c(t_{rcv} + t_{ei} + X_i/c_r)$ B) $c(t_{rcv} - t_{ei} - X_i/c_r)$ C) $c(t_{rcv} + t_{ei} - X_i/c_r)$ D) $c(-t_{rcv} + t_{ei} - X_i/c_r)$  Discipline: GNSS; Image Type: Diagram	In the provided DEM visualization, four areas (①, ②, ③, ④) are marked. Based on the terrain characteristics, which area is most likely to represent a valley? A. ① B. ② C. ③ D. ④  Modality: DEM; Image Type: DEM image	The image represents a remote sensing product derived from Landsat 8 bands using the equation $(\text{Band 5} - \text{Band 4}) / (\text{Band 5} + \text{Band 4} + 0.5) * 1.5$. Based on this index, which of the following statements is correct? A) The southern region has higher vegetation cover. B) The southern region experiences higher temperatures. C) The northern region has a lower density of buildings. D) The northern region has higher soil moisture.  Task: Environmental Analysis; Image Type: RS-Map

Figure 2. Examples from GeoMMBench, covering multiple disciplines, diverse sensor modalities, and a wide range of task types. Answering these questions requires advanced geoscience and RS expertise, along with specialized reasoning and interpretation skills.

studies, whereas the *test* set serves as the final benchmark for performance evaluation. This design ensures a standardized, transparent, and reproducible framework for assessing MLLMs in the geoscience and RS domain.

We highlight key dimensions of GeoMMBench that ensure a specialized and comprehensive evaluation (Fig. 1):

Disciplines: GeoMMBench encompasses four core disciplines: *RS*, *photogrammetry*, *GIS*, and *GNSS*. Advanced geospatial interpretation requires integrating knowledge across these domains. These disciplines are further supported by foundational subjects such as mathematics, physics, geography, cartography, and spectroscopy. The questions frequently integrate multiple disciplinary perspectives to ensure a holistic assessment.

Sensor modalities: The benchmark includes a broad range of sensing modalities, such as *MSI/HSI*, *SAR*, *LiDAR*, *DEM*, *optical* and *thermal* imagery. This diversity enables rigorous evaluation of a model’s ability to integrate multi-sensor data for complex Earth observation and geospatial analysis, which is essential for achieving expert-level intelligence.

Task spectrum: GeoMMBench employs a structured task hierarchy to systematically assess key geospatial competencies. It spans four levels: 1) *Theoretical knowledge* across multiple disciplines and subjects, forming the foundation for downstream perception and application tasks. 2) *Low-level perception*, covering image quality assessment and

RS-specific corrections (e.g., atmospheric, radiometric, and geometric corrections) for data preprocessing and quality control. 3) *Mid-level geospatial recognition* across sensor modalities, including scene/object recognition, geospatial relationship analysis, and temporal change detection. 4) *High-level geospatial applications* across diverse domains, such as environmental monitoring, agriculture, economic analysis, and RS mapping products.

Image formats: GeoMMBench incorporates diverse geospatial image formats, including *raw sensor imagery*, *maps*, *diagrams*, *tables*, *plots*, and various *RS mapping products*. This diversity challenges models to process heterogeneous visual data while integrating domain-specific knowledge to interpret textual and visual information and apply reasoning skills.

Together, these dimensions ensure that GeoMMBench provides a rigorous and holistic assessment of Expert AGI in geoscience and RS. Fig. 2 presents example questions covering various evaluation dimensions.

3.2. Benchmark Construction

Data collection. A team of geoscience and RS experts, including the co-authors, first defined the key evaluation dimensions encompassing disciplines, sensor modalities, and task categories. They then systematically outlined essential knowledge points within these dimensions. Experts

Benchmarks	Disciplines				Sensor Modalities						Task Spectrum					
	RS	Ph.	GIS	GNSS	Optical	HSI	SAR	DEM	LiDAR	Thermal	Pri.	Per.	Spa.	Qua.	Tim.	App.
VLEO-Bench [54]	✓	–	–	–	✓	–	–	–	–	–	–	✓	–	–	✓	–
EarthVQA [43]	✓	–	–	–	✓	–	–	–	–	–	–	✓	–	–	✓	–
GeoChat-Bench [19]	✓	–	–	–	✓	–	–	–	–	–	–	✓	✓	–	–	–
VRSBench [22]	✓	–	–	–	✓	–	–	–	–	–	–	✓	✓	–	–	–
TeoChat-Bench [18]	✓	–	–	–	✓	–	–	–	–	–	–	✓	–	–	✓	–
LHRS-Bench [32]	✓	–	–	–	✓	✓	–	–	–	–	–	✓	✓	–	–	–
GeoText-1652 [7]	✓	–	–	–	✓	–	–	–	–	–	–	✓	–	–	–	–
GeoBench-VLM [8]	✓	–	–	–	✓	–	✓	–	–	–	–	✓	✓	–	✓	–
XLRS-Bench [42]	✓	–	–	–	✓	–	–	–	–	–	–	✓	✓	–	–	–
CHOICE [3]	✓	–	–	–	✓	–	–	–	–	–	–	✓	✓	✓	–	–
GeoMMBench	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 1. Comparison of geoscience and RS benchmarks for MLLMs across key evaluation dimensions. Full names of all dimensions are listed in the Appendix.

were assigned specific knowledge points to develop multimodal questions using major textbooks, online resources, and academic literature, ensuring comprehensive coverage of geospatial expertise by creating new questions when needed. The process strictly adhered to copyright and licensing regulations, ensuring that no data was sourced from restricted materials.

Review process. All questions underwent a rigorous peer review by the expert team. The review process systematically identified and addressed two key issues: 1) Text-only inference bias: Some questions could be answered correctly without relying on the image input, thereby failing to assess true multimodal understanding. These questions were either revised to reinforce image dependency or excluded from the benchmark. 2) Incorrect samples: Questions with ambiguous phrasing, misleading answer choices, or incorrect solutions were identified and refined to uphold benchmark integrity, ensuring that all questions accurately reflected geospatial expertise and aligned with the benchmark’s objectives.

3.3. Comparisons with Existing Benchmarks

We elaborate the key differences between GeoMMBench and existing benchmarks from the following perspectives:

Breadth. Table 1 summarizes the evaluation dimensions of GeoMMBench relative to prior benchmarks. Most existing benchmarks are limited in scope, focusing primarily on perception-oriented tasks within the optical domain. GeoMMBench substantially expands this scope by integrating multiple disciplines, diverse sensor modalities, and a comprehensive task spectrum. This broader coverage enables more holistic evaluation of MLLMs and supports progress toward expert-level multimodal intelligence in geoscience.

Depth. Even when GeoMMBench shares evaluation dimensions with prior benchmarks, its questions demand deeper domain-specific reasoning. As illustrated in Fig. 2 (row 3, col. 3), GeoMMBench assesses spatial relationship recognition within authentic geoscientific contexts. While earlier benchmarks [8, 19] may include simple spatial queries such as identifying “left” or “right,” GeoMM-

Bench challenges models to determine relative directions using a compass or compute absolute distances from scale bars, which requires advanced spatial reasoning. Similarly, in RS application tasks, models must move beyond basic color recognition in thematic maps to the correct interpretation and use of spectral indices. For instance, GeoMMBench evaluates whether a model can interpret the SAVI* and apply it appropriately in a given context (row 4, col. 3). These tasks reflect a substantially deeper level of reasoning that combines conceptual understanding with analytical and application skills, extending well beyond traditional perception-based evaluation.

GeoMMBench remains **complementary** to existing benchmarks. Prior efforts provide valuable evaluations for optical imagery and foundational vision tasks that support many downstream geospatial applications. Rather than duplicating these contributions, GeoMMBench expands the evaluation landscape toward higher-level reasoning and interpretation skills that are essential for multimodal expert intelligence in geoscience and RS. Additionally, GeoMMBench is **expert-crafted**, differing fundamentally from prior benchmarks that repurpose perception datasets with text annotations or MLLM-generated questions. Each question is authored by geoscience and RS experts based on domain knowledge and authoritative references, ensuring methodological rigor and substantive depth. This expert-driven construction makes GeoMMBench both uniquely challenging and complementary to existing resources.

4. GeoMMAgent

Given the complexity and diversity of geoscience and RS tasks, it is challenging for a single MLLM to meet all domain requirements. A promising solution is to leverage intelligent agent systems that integrate general and specialized tools for adaptive problem-solving. To this end, we introduce GeoMMAgent, a multi-agent orchestration framework designed to enable expert-level interpretation for multimodal tasks in geoscience and RS.

*SAVI: Soil-Adjusted Vegetation Index

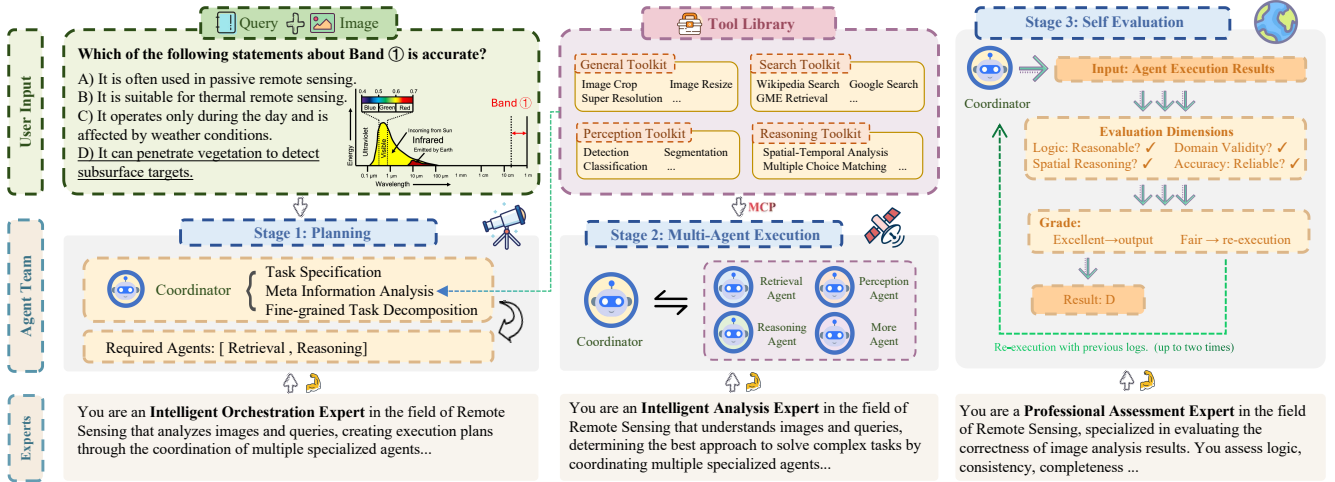


Figure 3. Overview of GeoMMAgent, a multi-agent framework that plans, executes, and self-evaluates multimodal tasks for expert-level geospatial understanding.

4.1. Overall Framework

As illustrated in Fig. 3, GeoMMAgent follows a *plan–execute–evaluate* paradigm comprising three stages: *Planning*, *Multi-Agent Execution*, and *Self-Evaluation*. Given an image–query pair as input, the system dynamically decomposes complex problems into structured subgoals, assigns them to specialized agents, and integrates their outputs into a coherent final answer.

Planning. A unified coordinator first interprets the input using an internal planner that formulates an executable plan specifying subgoals, required tools or agents, and scheduling strategies for execution order and dependencies.

Multi-Agent Execution. Based on the execution plan, the coordinator dispatches subtasks to specialized agents responsible for retrieval, perception, and reasoning. Each agent operates with domain-specific tools, such as object detection or geospatial reasoning models, and returns structured outputs. A comprehensive tool library comprising these specialized agents is described in Section 4.2. The coordinator then aggregates the results into an interpretable, evidence-based response.

Self-Evaluation. Finally, a self-evaluation module reviews the reasoning trace and generated answer, verifying consistency and correctness against the original plan. If inconsistencies are detected, the module triggers selective re-execution or refinement.

This modular and iterative workflow enables GeoMMAgent to perform transparent, interpretable, and expert-level analyses for multimodal geoscience and RS tasks.

4.2. Specialized Tool Library and Core Capabilities

We define three core geospatial capabilities that underpin expert-level intelligence: *knowledge*, *perception*, and *rea-*

soning. These capabilities reflect human expert cognition, as geospatial analysis is inherently complex, dynamic, and multidimensional. Accurate interpretation requires integrating information across multiple sensors, scales, and disciplinary contexts. To operationalize these capabilities, GeoMMAgent provides a tool library that equips agents with domain-specific functionalities for robust interpretation.

Knowledge tools handle the retrieval and synthesis of external information. We integrate multiple search engines (e.g., Wikipedia and Google) for general knowledge acquisition and adopt GME [57] for multimodal retrieval to match collected image–text information with the input image–query pair. Retrieved evidence snippets are returned with provenance, ensuring fact-grounded reasoning and expert-level explanations.

Perception tools include traditional RS models such as object detection, scene classification, and land use and land cover segmentation, trained with supervised learning on domain datasets. These models are optimized for geospatial perception tasks and provide accurate, interpretable results.

Reasoning tools consist of advanced MLLMs that generate contextual insights or intermediate reasoning steps from image–text pairs to support final answer generation. This module is automatically invoked when the coordinator identifies that complex logical structures must be resolved. In our implementation, we employ Qwen-VL-Max, though the framework can accommodate alternative reasoning models.

All tools are implemented in compliance with the Model Context Protocol (MCP) [14], ensuring scalability and modularity. This design allows new domain-specific tools to be seamlessly integrated, enabling GeoMMAgent to coordinate heterogeneous agents and perform transparent, expert-level multimodal reasoning in geoscience and RS.

5. Experiments

5.1. Baselines

Models. The evaluated models are grouped into three categories. 1) Closed-source MLLMs [9, 33–35, 39] released by commercial providers. 2) Open-source MLLMs further divided into general-purpose [1, 5, 10–12, 21, 23, 24, 26, 40, 44, 45, 56] and geoscience-specialized variants [18, 19, 32, 37]. 3) LLMs as text-only baselines [2] used to assess the contribution of visual inputs.

Human experts. To establish a human expert baseline, we invited participants with at least a master’s degree in a relevant field to complete the validation tests. They had access to course materials and personal notes but were strictly prohibited from using online search engines or AI assistants.

Evaluation. All models, including GeoMMAgent, are tested under a zero-shot setting without fine-tuning or additional supervision. Each model follows its default multiple-choice question-answering prompt for consistency. Model responses are evaluated through an automated rule-based pipeline using regular expressions to extract option letters and numerical answers, with invalid outputs marked as incorrect. Micro-averaged accuracy serves as the primary metric. All evaluations are conducted using the LMMs-Eval toolkit [55], with random-choice results reported as a baseline. Experiments are performed on NVIDIA A100 GPUs.

5.2. Main Results

We present the evaluation results in Table 2 and summarize our key findings as follows:

GeoMMBench poses significant challenges for current models. Human experts achieve a validation accuracy of 86.5%, outperforming all foundation models by a large margin. This pronounced gap underscores the limitations of existing models in geoscience and RS, revealing weaknesses in expert-level reasoning, multimodal integration, and domain-specific interpretation. These results demonstrate the rigor of GeoMMBench and its effectiveness in evaluating advanced geospatial competencies.

GeoMMAgent demonstrates superior capability. GeoMMAgent attains 88.4% accuracy on the test set, surpassing all MLLMs by substantial margins. It also achieves the same score by human expert on val set. This is due to its multi-agent design, enabling the intelligent decomposition of complex RS tasks and coordinated use of specialized tools, highlighting the great potential of agent-based systems for expert-level multimodal analysis in RS.

Open-source MLLMs are rapidly improving. State-of-the-art closed-source models still lead overall, with Gemini-1.5 Pro [39] reaching 70.7% accuracy. The leading open-source model, Qwen3-VL-30B [41], achieves 66.7%, narrowing the gap and even outperforming GPT-4o [34]. Given the importance of geoscience and RS for public and

sensitive applications, continued progress in open-source MLLMs is critical for accessibility and private deployment.

Geospatial-specific models require stronger instruction tuning. Domain-specialized MLLMs such as GeoChat [19], TeoChat [18], LHRS-Bot [32], and VHM [37] perform markedly below advanced general-purpose models despite fine-tuning on RS data. Their narrow focus on perception tasks limits reasoning and generalization in broader geospatial contexts. These results highlight the need for more diverse datasets and instruction-tuning strategies to enhance domain-specific expertise.

5.2.1. Error Analysis

To better understand the limitations of SOTA MLLMs in geoscience and RS, we analyze misinterpreted samples from Gemini-1.5 Pro and GPT-4o, the top-performing models from Google and OpenAI, respectively, as reported in Table 2. Fig. 4 presents representative error cases specific to geospatial tasks, highlighting common geo-interpretation failure patterns in model predictions. These cases underscore the need for domain-specialized advancements in future developments.

Visual-linguistic misalignment. A key challenge in evaluating Expert AGI for geoscience and RS is distinguishing between genuine gaps in domain knowledge and failures caused by misalignment between language and vision components. Our analysis reveals that while SOTA LLMs often exhibit strong geospatial knowledge in text-based tasks, their visual understanding remains inadequate, resulting in incorrect predictions. Fig. 4 (top-left) illustrates two representative cases. When given text-only input, GPT-4o and Gemini-1.5 Pro accurately describe the spectral characteristics of 1mm–1nm wavelengths (i.e., microwave) and recognize their ability to penetrate vegetation for subsurface detection. However, when required to identify the wavelength range for Band ① in a standard electromagnetic spectrum diagram, both models fail, selecting incorrect options despite possessing the necessary textual knowledge. A similar issue arises with Along-Track Scanning (Pushbroom) sensors. Both models correctly explain their advantages in a text-based setting but fail to recognize them visually, leading to misinterpretation.

These findings highlight a key limitation: while MLLMs may exhibit rich geospatial knowledge in text-based tasks, their visual encoders remain poorly aligned, preventing effective integration of visual and textual reasoning. Bridging this gap requires improving visual processing architectures to enhance the interpretation of domain-specific imagery in geoscience and RS.

Misinterpretation of geospatial relationships. Interpreting spatial relationships is fundamental in geoscience and RS, often requiring expertise in analyzing spatial elements such as compasses (for direction), scale bars (for distance), and latitude/longitude grids (for positioning). GeoMM-

Model	Val	Test	Results by Disciplines				Results by Sensor Modalities						Results by Task Spectrum					
			RS	Pho.	GIS	GNS.	Opt.	DEM	SAR	HSI	LiD.	The.	Pri.	Per.	Spa.	Qua.	Tim.	App.
Random	20.8	25.2	25.3	25.2	25.0	22.4	27.2	20.4	24.5	35.1	16.1	32.1	23.2	23.7	35.1	42.3	28.3	26.3
Human Expert	86.5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
<i>LLMs: Only Text as Input</i>																		
GPT-4 Text [2]	24.3	31.2	27.4	49.6	36.4	45.9	19.9	29.9	26.4	32.4	35.5	35.7	30.8	27.1	16.2	7.7	30.4	30.5
<i>Open-source MLLMs</i>																		
GeoChat [19]	21.6	35.4	34.1	40.9	34.1	34.1	41.7	22.8	34.6	27.0	46.8	21.4	31.1	36.8	29.7	26.9	30.4	33.9
TeoChat [18]	24.3	31.1	28.2	39.4	52.3	37.6	29.6	22.8	30.8	32.4	25.8	25.0	29.2	26.4	27.0	23.1	32.6	28.0
SkySenseGPT [29]	18.9	36.5	35.8	40.2	36.4	37.6	48.1	25.7	31.4	37.8	46.8	25.0	30.3	42.5	37.8	26.9	28.3	30.5
LHRS-Bot [32]	27.0	31.1	27.9	37.0	45.5	41.2	31.6	26.9	25.2	29.7	21.0	25.0	24.0	33.8	13.5	15.4	32.6	33.1
VHM [37]	37.8	39.1	39.0	40.2	27.3	43.5	56.3	28.1	30.8	35.1	41.9	35.7	27.7	46.2	35.1	38.5	41.3	44.9
LLaVA1.5-7B [23]	27.0	34.4	34.6	7.6	45.5	27.1	46.6	21.0	38.4	27.0	41.9	32.1	29.2	40.8	37.8	34.6	26.1	30.5
LLaVA1.5-13B [23]	29.7	38.2	38.9	32.3	50.0	27.1	57.3	27.5	37.7	32.4	32.3	28.6	32.4	45.5	45.9	34.6	41.3	35.6
LLaVA-NeXT-7B [24]	45.9	41.2	42.0	40.2	54.5	35.3	54.9	31.7	37.1	32.4	40.3	53.6	36.8	47.8	51.4	34.6	39.1	39.8
LLaVA-NeXT-34B [24]	59.5	54.2	52.8	63.0	56.8	60.0	60.2	42.5	52.2	51.4	74.2	35.7	50.7	51.8	59.5	46.2	54.3	55.9
LLaVA-OneVision-0.5B [21]	32.4	40.3	39.9	44.1	40.9	35.3	48.5	32.3	35.8	35.1	51.6	32.1	35.0	41.8	35.1	46.2	37.0	41.5
LLaVA-OneVision-7B [21]	56.8	57.3	55.7	64.6	65.9	64.7	66.5	50.3	47.8	59.5	62.9	46.4	53.5	57.5	54.1	65.4	52.2	51.7
CogVLM2 [45]	37.8	54.1	52.8	55.9	54.5	60.0	67.0	48.5	43.4	62.2	61.3	53.6	47.0	57.9	40.5	57.7	54.3	54.2
Mini-Monkey [17]	54.1	41.8	41.4	37.0	52.3	43.5	48.1	34.7	36.5	43.2	50.0	32.1	37.1	43.8	37.8	42.3	45.7	43.2
LLaMA3.2-11B [12]	35.1	40.1	38.1	46.5	50.0	43.5	46.1	32.9	35.8	43.2	45.2	28.6	37.1	40.1	29.7	42.3	30.4	36.4
InternLM-XComposer-2.5-7B [56]	37.8	46.2	45.3	47.2	63.6	45.9	54.4	37.7	37.7	70.3	48.4	50.0	39.9	48.5	35.1	46.2	54.3	48.3
InternLM-XComposer2-4KHD [10]	44.5	31.3	31.6	29.1	43.2	17.6	52.9	25.1	22.6	27.0	29.0	28.6	24.3	40.5	59.5	34.6	23.9	22.0
NVILA-8B [26]	43.2	45.3	42.4	57.5	47.7	55.3	34.5	45.5	37.7	67.6	56.5	46.4	42.8	39.5	35.1	30.8	45.7	51.7
NVILA-15B [26]	48.6	46.8	43.7	62.2	47.7	56.5	30.6	47.3	49.7	56.8	59.7	50.0	48.3	35.8	37.8	42.3	45.7	57.6
Phi-3.5v-8B [5]	59.5	56.8	55.2	61.4	65.9	62.4	65.0	46.1	42.1	59.5	64.5	53.6	50.4	55.2	54.1	57.7	60.9	64.4
Phi-3v-128k [1]	45.9	46.9	45.2	51.2	63.6	51.8	51.5	37.1	44.0	51.4	54.8	39.3	44.1	44.8	37.8	38.5	39.1	47.5
Phi-3.5v [1]	51.4	48.7	46.8	55.1	63.6	60.0	50.5	39.5	40.3	54.1	54.8	46.4	46.2	44.5	54.1	34.6	47.8	47.5
Phi-4-Multimodal [11]	48.6	55.3	54.0	63.8	61.4	62.4	58.3	50.3	47.2	70.3	61.3	50.0	53.3	53.2	40.5	65.4	47.8	58.5
Qwen2-VL-2B [44]	29.7	48.9	48.2	48.8	59.1	49.4	61.7	39.5	39.0	54.1	53.2	42.9	43.9	49.2	51.4	61.5	47.8	47.5
Qwen2-VL-7B [44]	51.4	59.8	59.1	56.7	70.5	61.2	69.4	51.5	54.1	67.6	67.7	50.0	53.8	63.5	62.2	46.2	58.7	61.9
Qwen2.5-VL-3B [40]	54.1	58.1	56.8	60.6	61.4	63.5	64.1	48.5	52.8	62.2	62.9	60.7	54.3	57.9	59.5	42.3	52.2	57.6
Qwen2.5-VL-7B [40]	56.8	62.5	60.6	68.5	72.7	70.6	66.0	52.1	55.3	70.3	74.2	57.1	59.8	59.2	54.1	50.0	52.2	64.4
Qwen3-VL-4B [41]	56.8	63.2	61.2	72.4	75.0	68.2	73.8	49.7	52.2	73.0	75.8	60.7	58.2	58.9	56.8	53.8	58.7	64.4
Qwen3-VL-8B [41]	56.8	65.8	64.0	70.1	77.3	70.6	72.8	59.3	55.3	75.7	75.8	67.9	61.4	62.5	51.4	57.7	60.9	71.2
Qwen3-VL-30B [41]	70.3	66.7	64.5	74.0	72.7	76.5	71.4	59.9	58.5	70.3	80.6	60.7	63.2	61.2	48.6	65.4	63.0	67.8
<i>Closed-source MLLMs</i>																		
GPT-4v [33]	56.8	61.0	59.3	71.7	61.4	67.1	63.6	61.7	54.1	45.9	71.0	46.4	59.8	58.5	37.8	53.8	56.5	69.5
GPT-4o [34]	56.8	68.5	65.6	78.7	77.3	80.0	71.4	62.3	61.0	64.9	80.6	67.9	65.8	63.9	48.6	69.2	56.5	72.0
GPT-o1 [35]	59.5	65.8	63.5	78.7	65.9	72.9	71.4	63.5	56.0	59.5	75.8	67.9	64.2	63.5	54.1	50.0	56.5	64.4
GPT-5 [36]	35.1	46.1	41.2	64.6	54.5	67.1	21.8	46.1	39.6	48.6	61.3	28.6	46.0	33.8	21.6	30.8	50.0	59.3
Gemini-1.5-pro [39]	64.8	70.7	68.9	76.4	86.4	72.9	74.3	64.7	67.3	64.9	77.4	75.0	71.5	68.2	56.8	61.5	58.7	62.7
Gemini-2.0-flash [9]	62.2	70.1	67.5	79.5	86.4	74.1	73.8	67.1	62.3	64.9	72.6	75.0	68.9	66.2	59.5	69.2	63.0	64.4
GeoMMAgent (ours)	86.5	88.4	87.6	89.8	93.2	97.6	78.2	94.0	91.2	89.2	74.2	89.3	92.2	82.9	97.3	76.9	97.8	83.1

Table 2. Evaluation results of MLLMs and text-only LLM baselines on the GeoMMBench *val* and *test* sets. The best-performing model in each category is highlighted in bold. Abbreviations: Full names of all dimensions and their subtasks are listed in the Appendix.

Bench evaluates these geospatial reasoning skills through a diverse set of targeted questions. As shown in Fig. 4 (upper right), existing MLLMs struggle with complex spatial reasoning, often failing to interpret relationships correctly, even when given clear visual cues such as legends. These findings highlight a critical gap in domain-specific spatial reasoning and emphasize the need for geospatial-aware MLLMs that can better integrate visual, numerical, and contextual information for precise spatial interpretation.

Sensor modality confusion. As shown in Fig. 4 (middle right), MLLMs may misinterpret images from different sensor types, failing to distinguish between various RS modalities. This challenge stems from training data bias, as most models are predominantly trained on RGB imagery, limiting their exposure to non-RGB RS such as HSI imagery. Overcoming this limitation requires expanding training datasets to include a more diverse and represen-

tative range of RS sensor modalities. This improvement will enhance models’ multimodal understanding, strengthening their ability to perform complex geospatial analyses and fostering synergistic multi-sensor processing.

Deficiencies in specialized geoscience disciplines. MLLMs often struggle with less common geoscience disciplines, likely due to their limited representation in training data. As shown in Fig. 4 (bottom left), the sample questions cover fundamental concepts in GIS and photogrammetry—topics considered standard knowledge in the field. However, even the two best-performing proprietary MLLMs fail to answer them correctly, revealing significant gaps in their domain understanding. This highlights a critical gap in existing models and underscores the need for specialized geospatial MLLMs with deeper domain knowledge to improve reasoning and interpretation in geoscience and RS.

Visual-Linguistic Misalignment		Geospatial Relation Misinterpretation	
Image-based question Which of the following statements about Band ① is accurate? Answer: A B	Equivalent Text-only question Which of the following statements about the spectral band in the 1 mm – 1 m wavelength range is accurate? Answer: D D Shared Options: A) It is often used in passive remote sensing. B) It is suitable for thermal remote sensing. C) It operates only during the day and is affected by weather conditions. D) It can penetrate vegetation to detect subsurface targets.	Question In which direction from the hospital (marked by red box) is the parking lot (blue box) located? Options A) East B) Northeast C) Northwest D) West Answer C B	Question What is the most likely actual distance between points A and B in the figure? Options A: 67 m B: 321 m C: 167 m D: 496 m Answer A C
Image-based question What is a major advantage of the remote sensing scanning method shown in the right panel? Answer: B B	Equivalent Text-only question What is a major advantage of the Along-track scanning (pushbroom)? Answer: C C Shared Options: A) Higher spectral resolution B) Higher spatial resolution C) More consistent data quality D) Wider range of spectral wavelengths	Sensor Modality Confusion Question Which panel in the image is most likely to be a hyperspectral image in remote sensing? Options A) Panel 2 B) Panel 3 C) Panel 7 D) Panel 1 Answer B B	
Specialized Discipline Deficiency Options A) Polygon, Point, Line B) Point, Line, Polygon C) Polygon, Line, Point D) Line, Polygon, Point Answer A B		Image Perception Failure Question Which remote sensing quality issue is present in the image? Options A) Twisted Object B) Spectral Overflow C) Stitching Misalignment D) Seam Line Answer C D	
Question This image shows azimuth, tilt, and swing rotation angles in Photogrammetry. What is the purpose of the x'y'z' coordinate system? Options A) To represent the photo measurement system aligned with the tilted photo plane. B) To serve as an intermediate coordinate system parallel to the ground coordinate system XYZ. C) To represent the nadir point projection. D) To simplify the calculation of azimuth, tilt, and swing angles Answer A A		Question What object is enclosed by the red bounding box in this image? Options A) Reach Stacker. B) Truck. C) Storage Tank. D) Shipping Container. Answer D D	

Figure 4. Representative error cases from advanced MLLMs specific to geospatial tasks. : GPT-4o; : Gemini-1.5 pro.

Image perception failure. While GeoMMBench does not primarily focus on perception tasks for raw RS images like prior benchmarks (as discussed in Section 3.3), it incorporates domain-specific recognition tasks that are essential for geoscience applications. Our evaluation indicates that while advanced MLLMs perform well in coarse-level recognition tasks for RGB images (e.g., scene classification), they struggle with more sophistic recognition tasks, as shown in Fig. 4 (bottom right). These challenges span from low-level assessments, such as RS-specific image quality inspection and correction, to mid-level tasks, including fine-grained object recognition. Since these recognition capabilities serve as the foundation for intelligent RS interpretation, improving MLLMs' ability to handle domain-specific geospatial perception tasks remains a crucial direction for future research and development in geospatial AI.

6. Conclusion

In this work, we introduced GeoMMBench, the first expert-crafted, multidisciplinary benchmark for evaluating MLLMs in geoscience and RS. Through comprehensive experiments, we revealed substantial challenges that current models face in expert-level reasoning, perception, and multimodal integration. To address these limitations, we developed GeoMMAgent, a multi-agent orchestration framework that integrates domain-specific tools for adaptive task planning and expert-level interpretation. GeoMMAgent achieves state-of-the-art performance on GeoMMBench, surpassing all existing models and approaching human expert performance. We believe that GeoMMBench and GeoMMAgent together lay a foundation for advancing intelligent, transparent, and domain-specialized multimodal agents in geoscience and beyond.

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